

FRUIT JUICE CLASSIFICATION USING IMPEDANCE AND PHASE MEASUREMENT CIRCUIT

**INTERDISCIPLINARY UNDERGRADUATE
PROJECT (UGQ301)**

PROJECT INSTRUCTOR

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INTRODUCTION

Fruit Juice type can be segregated using juice's unique sugar profile—the specific ratio of glucose, fructose, and sucrose—which serves as a identifying biomarker. Traditional methods for sugar profiling, such as High-Performance Liquid Chromatography (HPLC) or laboratory-grade Electrical Impedance Spectroscopy (EIS), are highly accurate but require expensive, bulky equipment and specialized training.

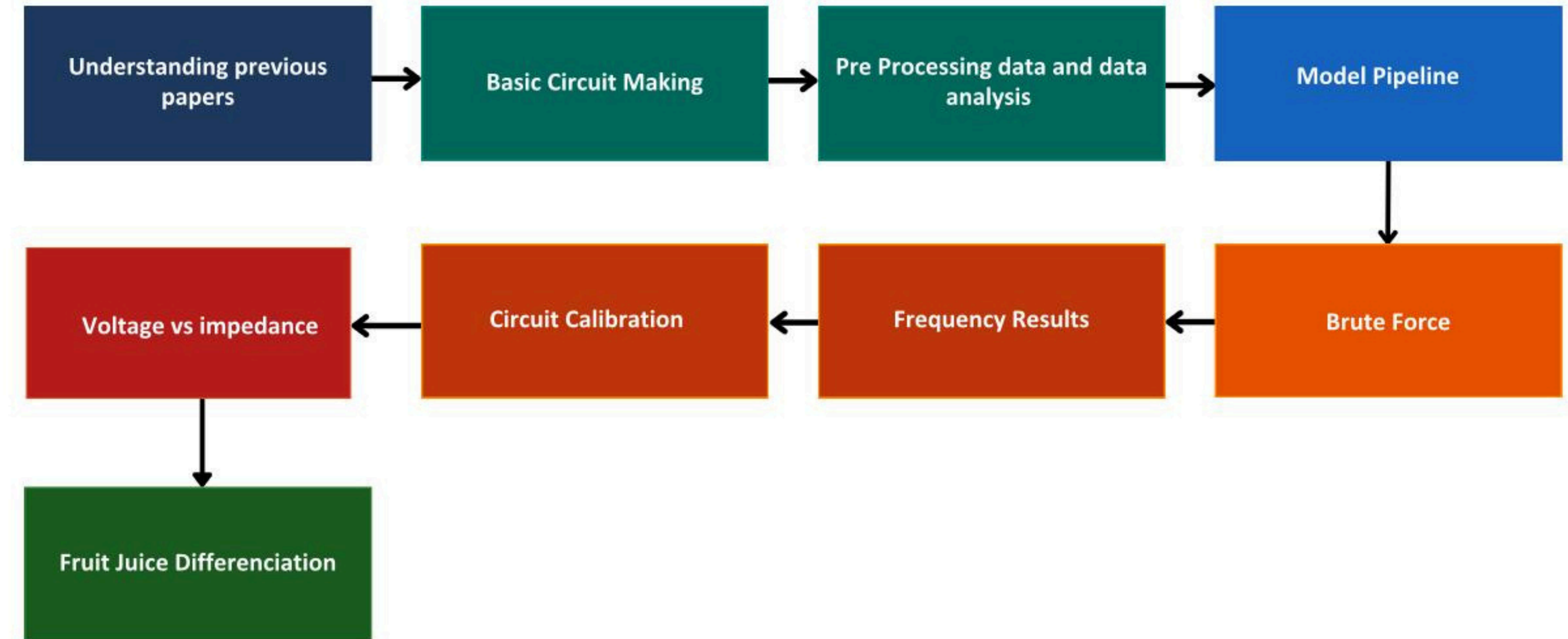
We have developed a custom Analogue Conductivity Meter.

we aim to provide a real-time classification system.

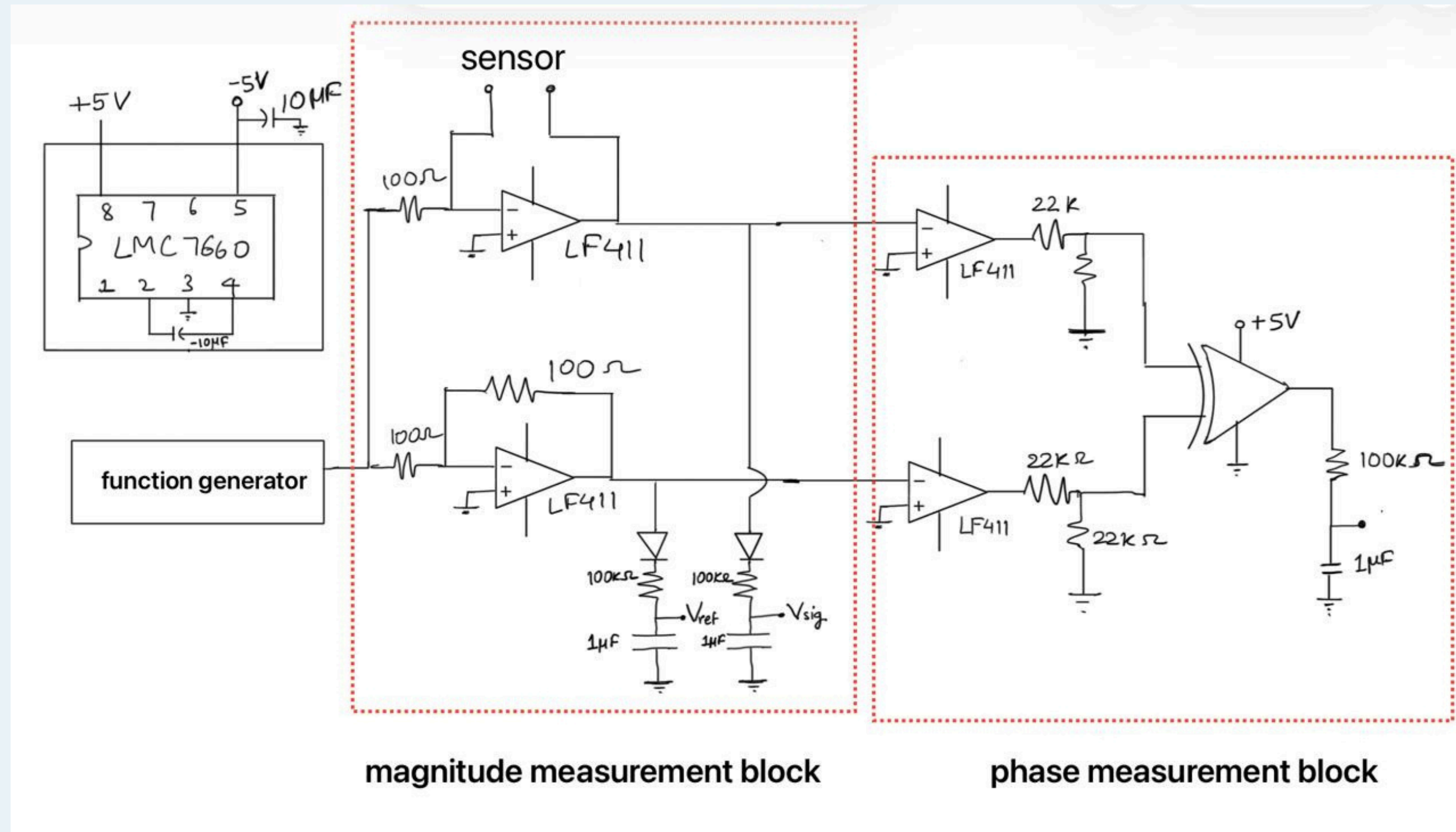
This report details the hardware design, optimal frequency, the empirical calibration of the voltage-to-impedance relationship, and the integration of these features into a microcontroller-based classifier to identify juice flavors with portable efficiency.

Flowchart

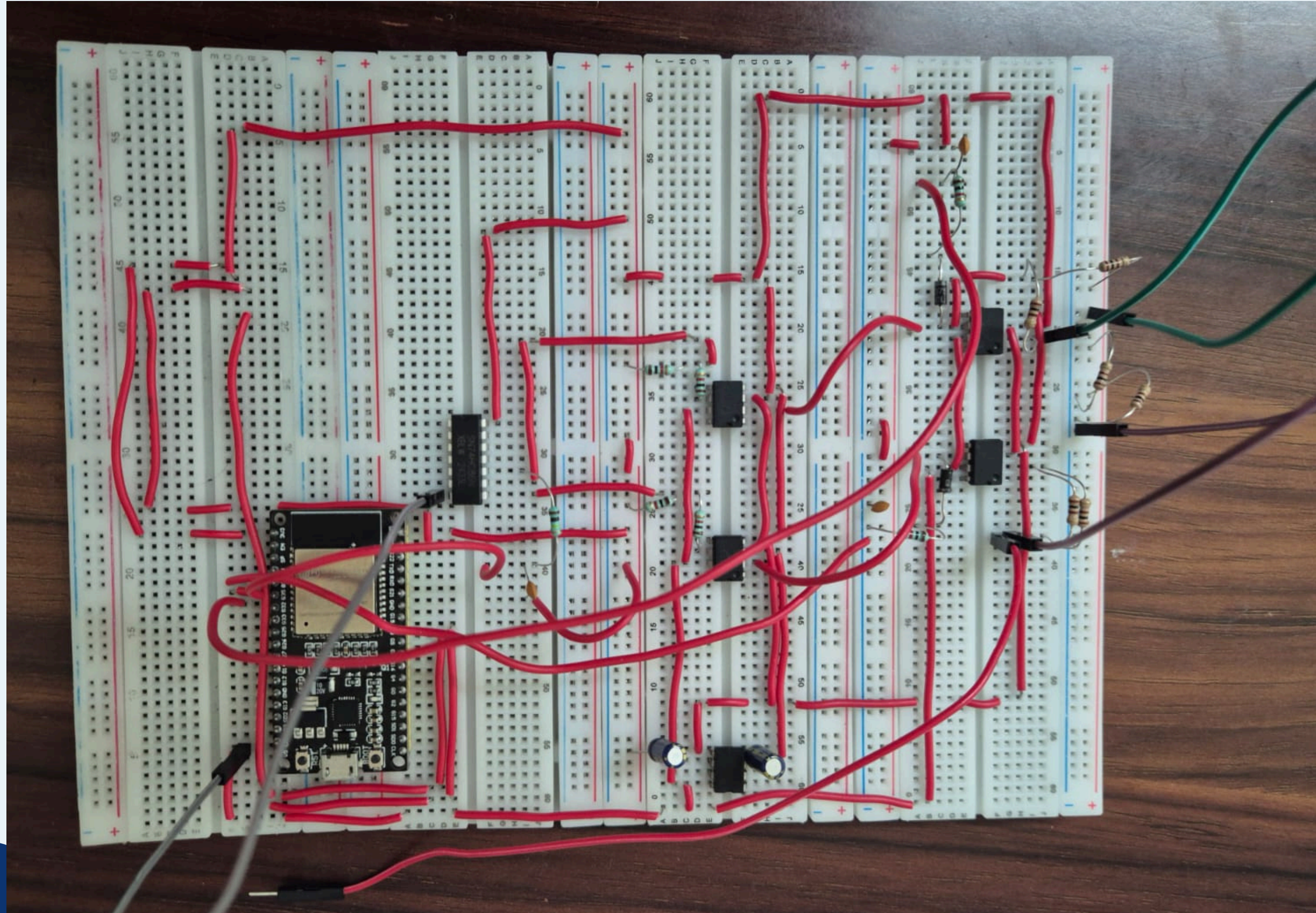
Step-by-Step Processing



CIRCUIT DIAGRAM



IMPLEMENTATION



DATASET DESCRIPTION

Dataset Description

juice_data.csv — Structure

Field	Description
Juice Name	Categorical juice type label
Sample ID	Unique integer per sample
Timestamp	t = 0, 2, 4 minutes
Z f1 – Z f50	Impedance magnitude (50 pts)
ϕ f1 – ϕ f50	Phase angle (50 pts)

5 Juice Classes

100 Features

Data Characteristics

- ⚠ Inconsistent column naming
- ⚠ Mixed data types in raw CSV
- ⚠ Missing values in some channels
- ⚠ Multiple timestamps requiring separation

→ Addressed in preprocessing pipeline

3 Timestamps

t=4 Used

Data Preprocessing Pipeline

1

Libraries

pandas & numpy
for data manipulation

2

Timestamp Selection

Retain $t = 4$ only
(lowest intra-class variance,
full electrode equilibration)

3

Feature Construction

Merge $|Z| + \phi$ at each freq
Feature vector $\in \mathbb{R}^{100}$

Feature Vector Formula

$$\mathbf{x} = [|Z|f1, \phi f1, |Z|f2, \phi f2, \dots, |Z|f50, \phi f50] \in \mathbb{R}^{100}$$

DATA ANALYSIS

What is LDA?

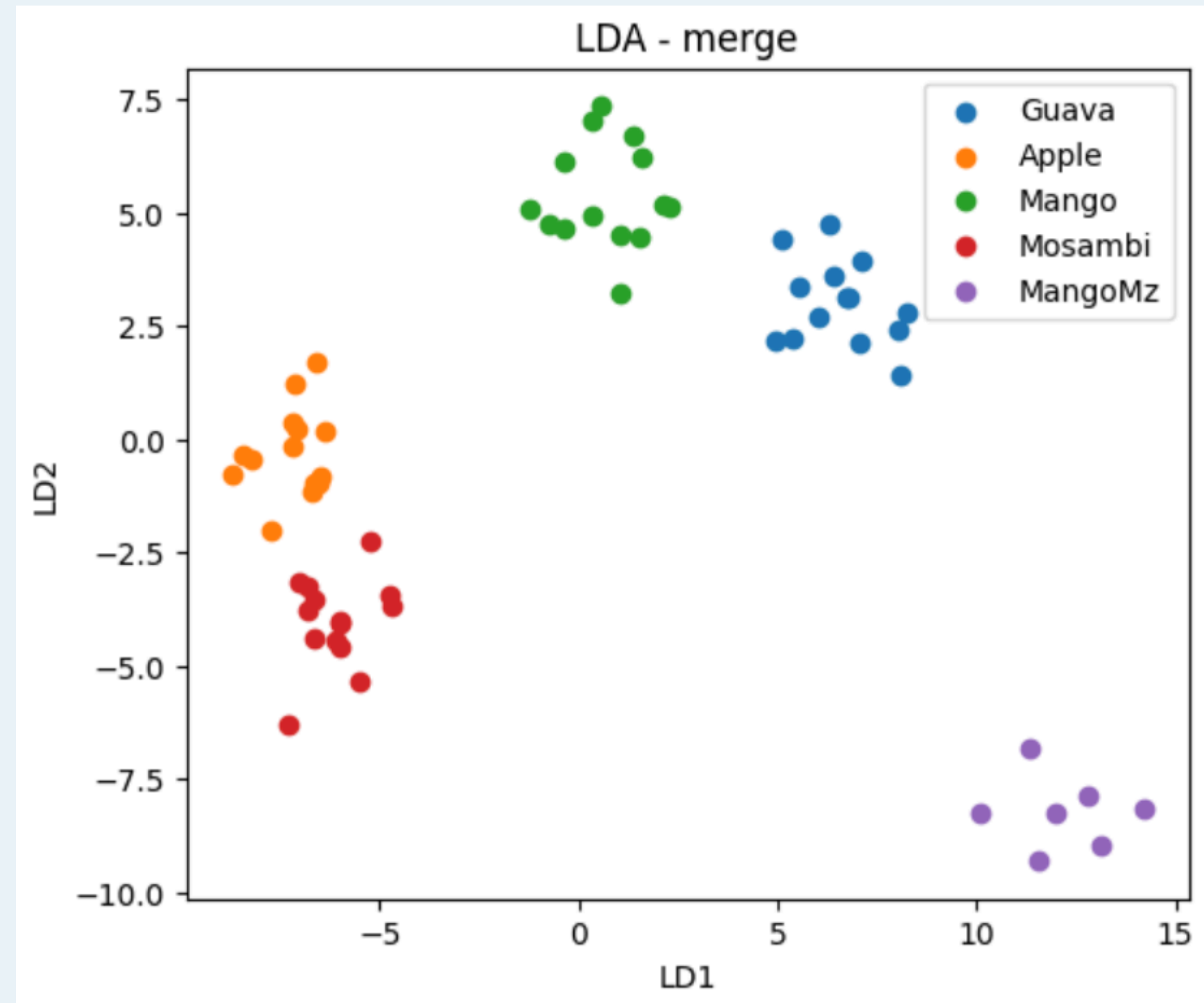
Linear Discriminant Analysis (LDA) reduces high-dimensional data to its most class-discriminative directions by maximising the Fisher criterion:

$$J(w) = w^T S_B w / w^T S_W w$$

where S_B = between-class scatter matrix and S_W = within-class scatter matrix.

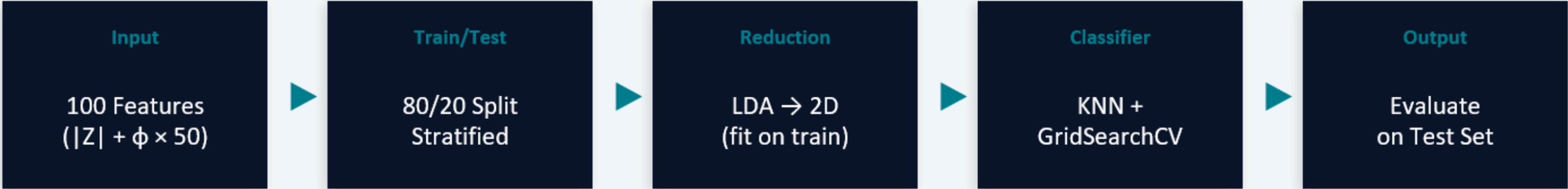
Findings from LDA Plot

- Strong separability for most juice classes
- Some overlap for similar ionic compositions
- Validates EIS as a discriminative modality
- Motivates LDA as a dimensionality reduction step



DATA ANALYSIS

Baseline Classification Pipeline: All 50 Frequencies



Hyperparameter Grid (GridSearchCV, 5-fold CV)

n_neighbors	[3, 5, 7, 9, 11]
weights	["uniform", "distance"]
metric	["euclidean", "manhattan"]

Baseline Results

74.0%

Test Accuracy

Pre-LDA Features: 100

LDA Dimensions: 2

CV Folds: 5

Optimal Frequency Selection — Motivation & Approaches

Why reduce to 2–3 frequencies? Measuring 50 frequencies is impractical — measurement time, circuit complexity, and power consumption all scale with frequency count.

Approaches Explored

PCA	t-SNE	ANOVA F-test	LDA Weights
⚠ Produces linear combos — no individual freq ranking	⚠ Non-parametric — cannot select specific inputs	⚠ Ignores feature interactions	⚠ Preliminary ranking — no optimal pair/triplet

✓ Chosen: Brute-Force Combinatorial Search

Evaluates ALL possible subsets of k frequencies through the full pipeline — guarantees the globally optimal subset. Accounts for both individual discriminability and frequency complementarity.

Brute-Force Search: Pairs vs Triplets

Pairs (k=2)

$C(50,2) = 1,225$ pairs evaluated

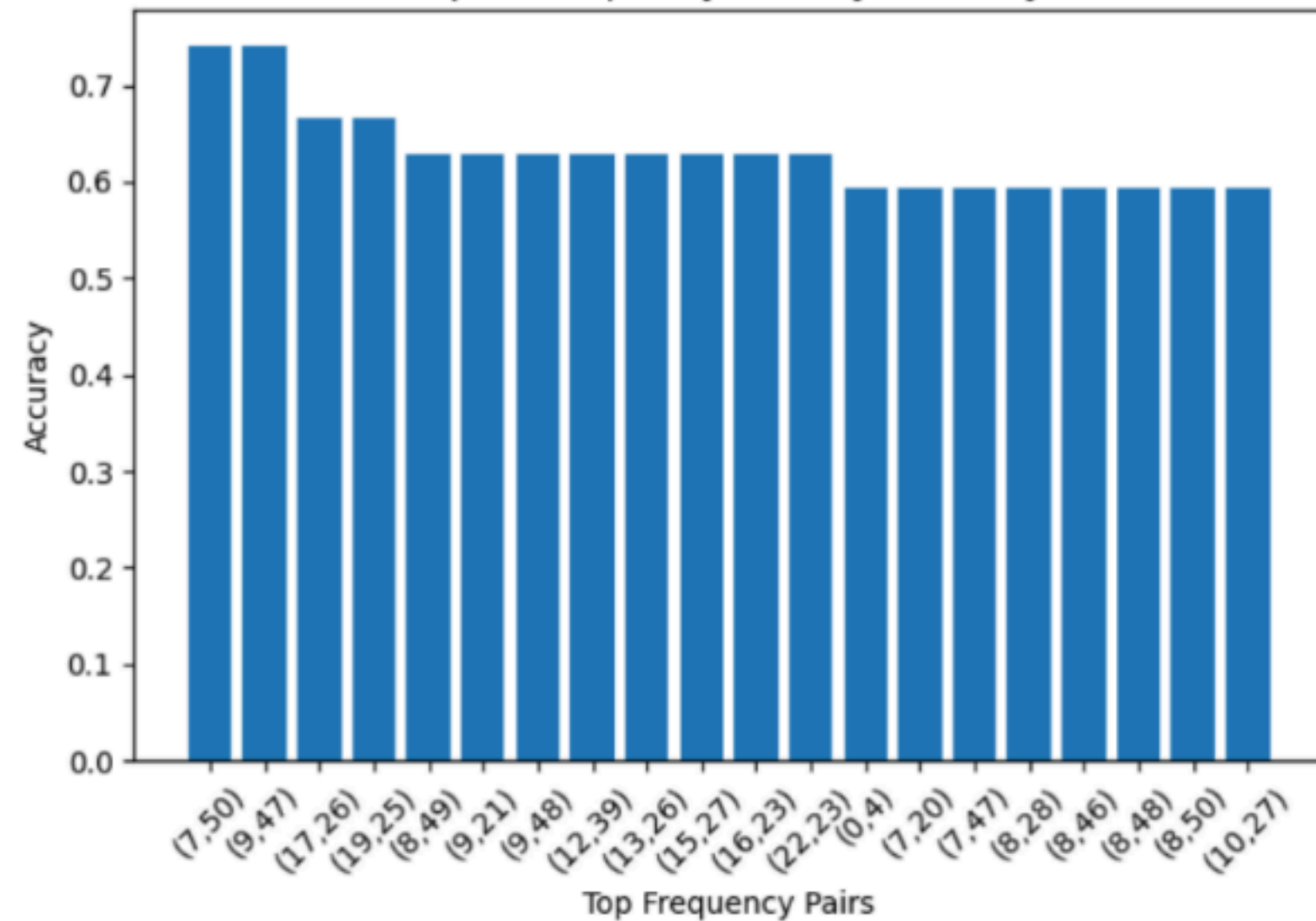
Best accuracy: ~74%

Triplets (k=3)

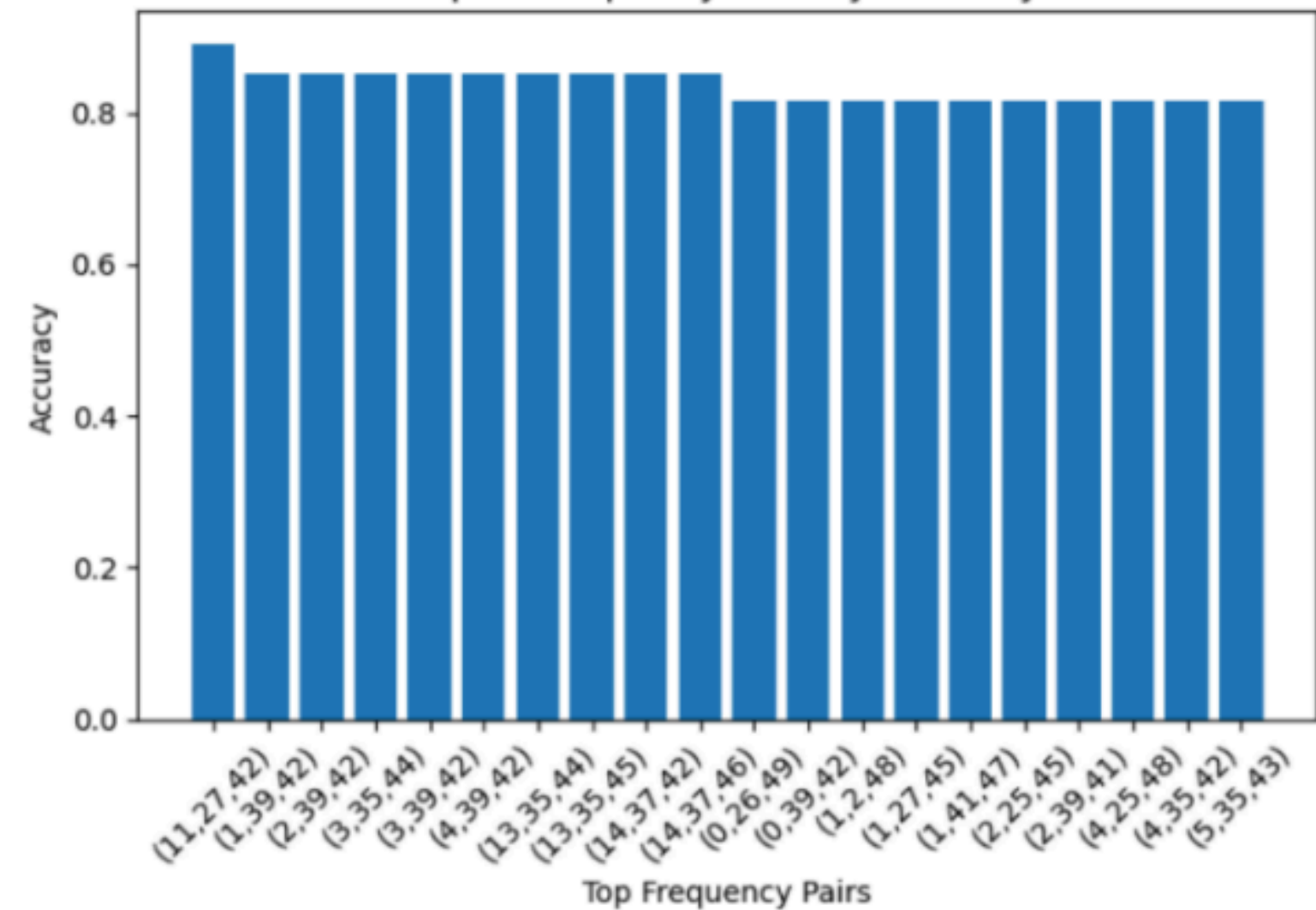
$C(50,3) = 19,600$ triplets evaluated

Best accuracy: 88.8%

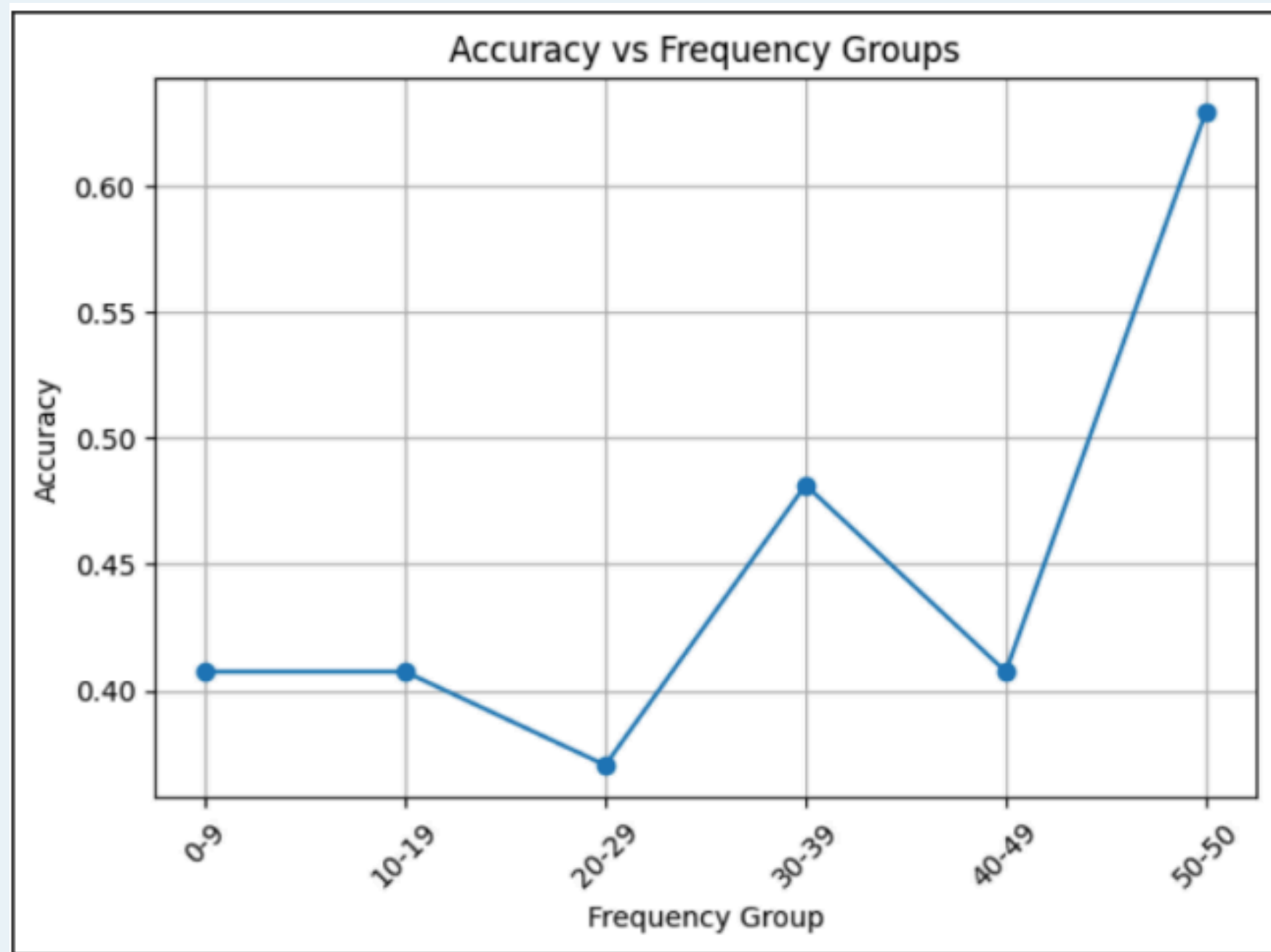
Top 20 Frequency Pairs by Accuracy



Top 20 Frequency Pairs by Accuracy



Sequential Group Analysis – Sanity Check



Final Accuracy: 0.48148148148148145

Classification Report:

	precision	recall	f1-score	support
Apple	0.57	0.67	0.62	6
Guava	0.40	0.67	0.50	6
Mango	0.67	0.67	0.67	6
MangoMz	1.00	0.33	0.50	3
Mosambi	0.00	0.00	0.00	6
accuracy			0.48	27
macro avg	0.53	0.47	0.46	27
weighted avg	0.48	0.48	0.45	27

Confusion Matrix:

```
[[4 2 0 0 0]
 [0 4 1 0 1]
 [0 2 4 0 0]
 [0 0 0 1 2]
 [3 2 1 0 0]]
```

Results: Best Frequency Triplets

From all the method the best accuracy of model was achieved of applying brute force for a group of 3. The brute-force search over all 19,600 triplets identified the optimal set of three frequencies.

```
=====
BEST PAIR FOUND
=====
Best Pair: (11, 27, 42)
Best Accuracy: 0.8889
```

```
... Final Accuracy: 0.8888888888888888

Classification Report:
              precision    recall  f1-score   support

   Apple      1.00      0.67      0.80         6
   Guava      1.00      0.83      0.91         6
   Mango      0.86      1.00      0.92         6
MangoMz      1.00      1.00      1.00         3
  Mosambi      0.75      1.00      0.86         6

 accuracy      0.89         27
  macro avg      0.92      0.90      0.90         27
  weighted avg      0.91      0.89      0.89         27

Confusion Matrix:
[[4 0 1 0 1]
 [0 5 0 0 1]
 [0 0 6 0 0]
 [0 0 0 3 0]
 [0 0 0 0 6]]
```

R and C values corresponding to min max frequency

Freq. point	Frequency (Hz)	ω (rad/s)	Type	$ Z $ (Ω)	Phase ($^\circ$)	R (Ω)	C
11	758.58	4766.28	MAX	59.24	-15.47	61.47	0.9447 μ F
11	758.58	4766.28	MIN	27.47	-25.31	30.39	3.2652 μ F
27	14454.4	90819.66	MAX	54.22	-1.29	54.23	4.572 nF
27	14454.4	90819.66	MIN	22.72	-2.3	22.74	19.45 nF
42	229086.77	1439394.6	MAX	53.96	-0.22	53.96	49.44 pF
42	229086.77	1439394.6	MIN	22.53	-0.16	22.53	86.11 pF

OUTPUT VOLTAGES

61.47 ohm, 3.3 uF → 0.87

60.3 ohm, 1uF → 0.46

freq=758.58 Hz						
	C=1 uF			C=3.3uF		
R(ohm)	Vsignal	Vref	Vphase	Vsignal	Vref	Vphase
36	0.51	0.72	0.16	0.47	0.72	0.16
41	0.55	0.72	0.16	0.49	0.72	0.16
46.2	0.64	0.72	0.17	0.52	0.72	0.16
55	0.8	0.72	0.19	0.53	0.72	0.16
50	0.75	0.72	0.16	0.57	0.72	0.16

Values of Z and Vsig/Vref

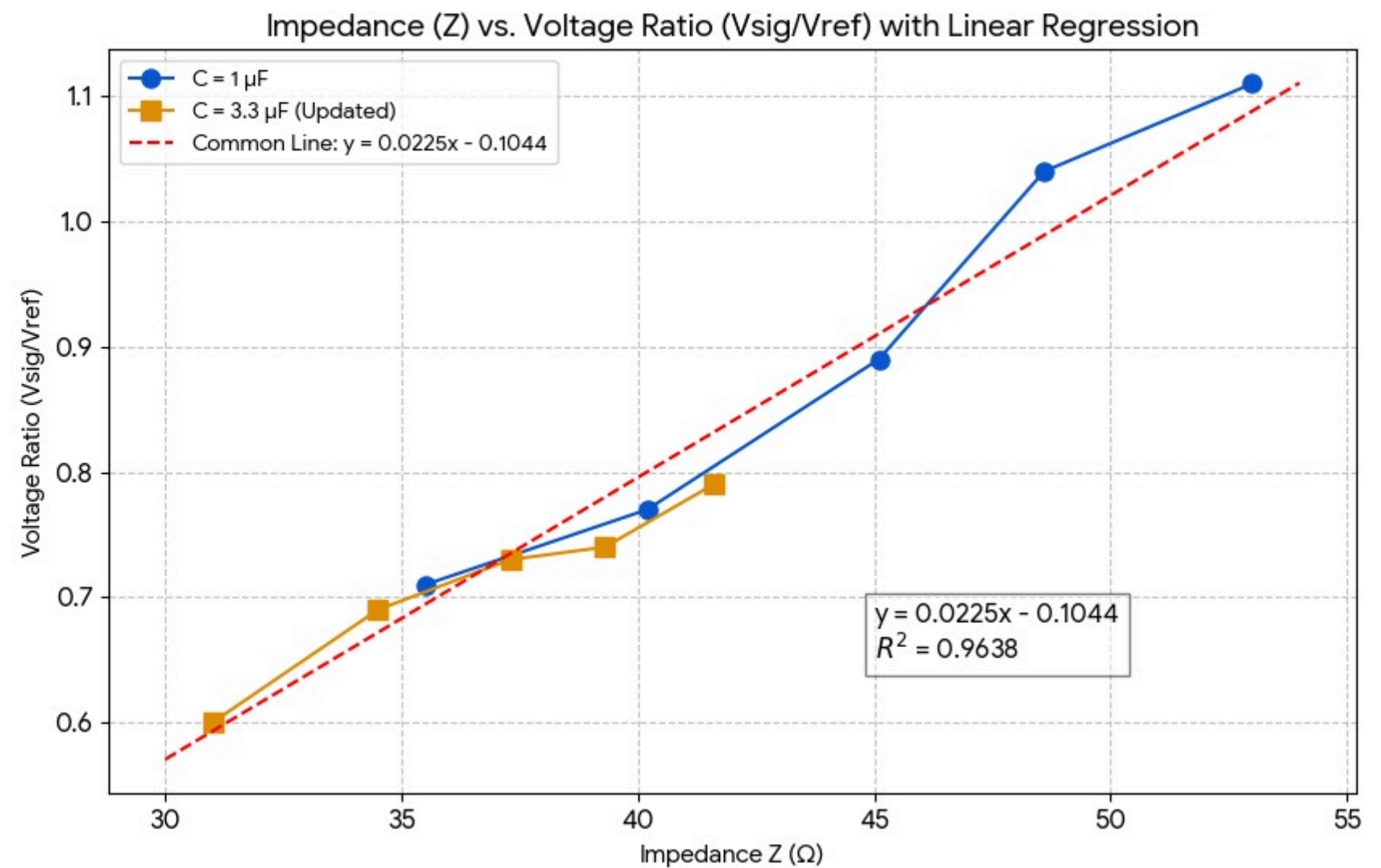
Frequency= 758.58 Hz	
C=1 uF	
Z	Vsig/Vref
35.5	0.708
40.2	0.764
45.1	0.889
48.6	1.042
53.1	1.111

Frequency= 758.58 Hz	
C = 3.3uF	
Z	Vsig/Vref
31.3	0.65125
34.5	0.68705
37.4	0.72868
39.3	0.73561
41.6	0.79242

IMPEDANCE VS VOLTAGE RATIO

From Combining these two curves through linear regression we got the line equation:

$$y=0.0225x-0.1044$$



THE FINAL STEP

From this line equation

$$y=0.0225x-0.1044$$

we got

$$x=44.45y+4.64$$

Using hydrogel sensor we can get voltage ratio and
determine the impedance

SPECIAL THANKS

DEBJIT SARKAR

DEBAKAR ROY



THANK YOU

